

Goal

- Goal is to go from audio to an insight stored in a database with a visualization that is repeatedly updated over a period of time
- Data needs to flow cleanly and routinely through the system given the large amount of moving parts
- Each DAG needs to work independently of the other without dependencies and accounting for potential failures

Architecture:

- The system should support pointing to 3 different environments (DEV-QA-PROD)
- There will be a single convo-set each linked to a single workspace for each environment that will contain a rolling window based on a number of files
- The state of files is maintained across BigQuery with the status field

DAG Breakdown:

1. Transcription DAG (TBD)
2. Ingestion DAG
3. Insight DAG

DAG 1 – Transcription DAG (TBD)

This DAG is responsible for taking the audio files in a GCS Bucket and converting them into transcribed CSVs. These CSVs then need to be pushed up into a "conversation" table in BigQuery.

Steps:

1. Pull down file names in audio files bucket
2. Compare against file names in the conversation table currently already transcribed
3. Download the audio files that need to be transcribed
4. Call the transcription provider and save CSV
5. Aggregate CSVs per call into HF Supported CSV format
6. Upload into BQ "conversations" table with the individual file names of the audio files to avoid re-transcription

Edge Case

- Over 1000 rows in a single id

Frequency:

- TBD

DAG 2 – Ingestion DAG

This DAG is responsible for ingesting the supported conversations from the “conversations” BQ table into a HumanFirst conversation set. The conversation set is to keep a rolling window of transcription files until the Insight DAG processes them. The conversations should be loaded with all the relevant metadata and a file name corresponding to the ingested file time. Ex: *“file-2024-12-04t053039.279879.csv”*.

Tasks:

1. check_unprocessed_conversations
2. validate_conversation_set_exists
3. fetch_conversations
4. upload_to_humanfirst

Steps:

- Check the “conversations” BQ table for the processed_status for count of UNPROCESSED records
 - If none found – DAG sleeps till next run since nothing to ingest
- Check if the conversation source ID (denoted in Variables) exists before pulling anything down
 - list of conversation sets from HumanFirst is checked to find src ID to push the file
 - If the convo_set was deleted or the source doesn’t exist, DAG will fail as expected
- If UNPROCESSED conversations exist, pull down X batch of records into a Pandas DataFrame format with the relevant metadata
 - Custom query is used here to CAST to UNIX_SECONDS timestamp from the BQ standard
 - Batch size should be configurable via Variables in Airflow (*INGESTION_BATCH_SIZE*)
 - Saves the dataframe to a temporary file for next task
- Create the file name corresponding to timestamp in iso format (*file-2024111T0000.csv*) and ensure that is included as metadata
- Make POST request to the end point and load the file into the conversation set
 - A mapper defined will map the CSV columns
 - Retrieve the triggerID from the response
- Poll triggerID to ensure the trigger completes successfully until max polling time is reached

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- End trigger can either be marked TRIGGER_STATUS: _FAILED, _COMPLETED, or _CANCELLED
- Perform rolling window check and delete the last file if over the file limit
- Update the processed_status in “conversations” BQ table to INGESTED status and the hf_filename column

Error Points:

- BQ doesn't respond with rows
- The configured conversation set source ID was not found
 - Request failed
 - Conversation Set was Deleted
- The conversation set POST request to upload failed
 - Request failed
 - Trigger failed

Logging:

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Improvements / Extra's:

- Implement backoff instead of static polling every 5 seconds to reduce API requests sent

Frequency:

- TBD

DAG 3 – Insight DAG

This DAG is responsible for generating the insights using the Pipelines on HumanFirst. The generated insights should make use of the cache and the rolling window of files. The insights are linked with a single unique FOREIGN KEY between the conversations table and the insight table. This DAG needs to handle key value pivot from the pipeline export for a defined schema. The DAG also needs to sequentially handle calling multiple pipelines for a single insight topic (ex: two pipelines split for calls/chats).

Steps:

- Query conversation files list for the conversation_src_id for the defined convo-set
- Query BQ to retrieve the last written insight to “insights” table for the filename processed
 - Map the last written file name and find which files are not yet processed AFTER that last written insight and store in memory for export

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- Edge case here is that files that don't pivot or don't have any insights generated for that pipeline will never exist in BQ – This should be handled by ALWAYS checking the last written file – If a file with no insights exists in the convo-set, it won't get reprocessed infinitely if the next file had some written insights
- For each pipelineID passed, trigger pipelineID in the workspace sequentially
 - Optional check if any pending triggers / running triggers in workspace (prob running only)
- Poll for trigger status for defined time until TRIGGER_STATUS_COMPLETED
 - If trigger status is _CANCELLED / _FAILED – retry logic should be delay X seconds and retry trigger with a maximum of 3 retries
- Once trigger completes - sleep for 30s-1m to allow for index to be ready for download
 - Ensure every pipelineID that was passed returned successfully otherwise fail DAG
- Query export pipeline filtered per filename in the list
 - Option 1 – Aggregate into a single DF after each query and upsert in one go to BQ – Need to perform one large pivot - Implemented
 - Failure case where if one file doesn't have the output correctly it might not pivot correctly (very unlikely) – Pipeline export file will remain consistent so shouldn't be a problem
 - Option 2 – Query and Upsert Per File – Need to perform multiple pivots ect
 - Failure case where only portion of the files get upserted – however the next DAG run will pickup since each file will be fully written or not written at all
- Upsert into BQ “insights” table & update the “conversations” table as PROCESSED
 - Run delete query before inserting to ensure that no duplicate convo-ids ever end up in insights table

Error Points:

- The configured conversation set source ID was not found
 - Request failed
 - Conversation Set was Deleted
- Pipeline fails or in cases of more than one, only one pipeline successfully completes despite retries
 - The DAG should FAIL and not upsert partial insights
- Index isn't ready after delay for exporting and export fails
 - Retry logic should pick up here
- Pivot fails for one of the files in multiple files
 - The only way the pivot fails on a key-value pipeline is if the index columns are missing (conversation_id, agent_id, .. ect) which is unlikely at this stage of the pipeline

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- If a row in the file didn't have any insight for that key, fillna will mark all blanks with None. Any extra keys via LLM hallucination will be removed.
- In this scenario if a pipeline completed but pivot doesn't work – we can imagine that the file is corrupted and doesn't have in the index columns

Logging:

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Improvements / Extra's:

- Batch predict optional step – currently not implemented

Frequency:

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Database Schema

“conversations-raw” Table

Field Name	Data Type	Mode	Notes
conversation_id	STRING	REQUIRED	
conv_created_at	TIMESTAMP	REQUIRED	
utterance	STRING	NULLABLE	
speaker	STRING	REQUIRED	
member_level	FLOAT	NULLABLE	
intent	STRING	NULLABLE	
agent_id	STRING	REQUIRED	
campaign_nm	STRING	NULLABLE	
sdr_success_ind	STRING	REQUIRED	
url	STRING	NULLABLE	
processed_status	STRING	REQUIRED	UNPROCESSED, INGESTED, PROCESSED, UNKNOWN
hf_conv_filename	STRING	NULLABLE	"file-2024111T0000"

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created_at	TIMESTAMP		Probably good idea to add a created at date independent of the conversation timestamp
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"insights" Table

[illegible]